

# Predictive Modelling of Parkinson's Disease Using Speech Biomarkers and Tremor-Based Features

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### Abstract

Parkinson's Disease (PD) is a progressive neurological disorder that affects movement, speech, and coordination. Early diagnosis is crucial for effective management; however, conventional clinical methods often lead to delayed detection. This study proposes a predictive approach utilizing voice biomarkers and hand tremor features.

Existing machine learning approaches, including Random Forest, Support Vector Machines, and basic Neural Networks, exhibit limitations such as overfitting, poor generalization on small datasets, and inadequate handling of multimodal data. Furthermore, deep learning models demand significant computational resources, limiting their practical applicability.

The advanced machine learning algorithms that will be used include Gradient Boosted Decision Trees, Random Forests, Extra Trees Classifier, and Deep Learning (i.e., Multi-Layer Perceptron). The data analysis will occur to determine patterns related to PD.

An ensemble Voting Classifier will be used to further enhance accuracy by combining multiple models. The study will also include feature importance analysis to identify key factors influencing PD prediction, supporting future research and early diagnosis efforts.

### Introduction

Parkinson's Disease (PD) is a progressive neurodegenerative disorder characterized by motor and non-motor impairments, where early diagnosis remains a significant challenge. In recent years, voice biomarkers have gained attention as a non-invasive and cost-effective diagnostic tool. However, several limitations persist in voice-based PD detection. Speech signals are highly sensitive to environmental noise, recording conditions, and patient variability, leading to inconsistent model performance [5]. Additionally, dysphonia-based features often lack robustness across diverse populations and recording devices, reducing generalizability [9]. Studies such as Little et al. (2009) demonstrated promising results using voice features for PD detection but highlighted issues related to dataset size and real-world applicability [26], while Sakar et al. (2016) reported high classification accuracy but noted challenges related to overfitting and interpretability [3]. Recent works (2025–2026) further emphasize challenges such as variability in acoustic parameters, lack of standardized datasets, and difficulty in interpreting voice-based predictions despite the use of advanced machine learning and explainable AI techniques [14].

Similarly, hand tremor analysis, another widely used biomarker for PD detection, also exhibits several disadvantages. Tremor-based approaches rely heavily on motion sensors, wearable devices, or image/video-based analysis, which may not always be accessible or affordable [18]. Moreover, tremor severity can vary significantly depending on patient conditions, medication, and activity, making it difficult to obtain consistent measurements [17]. Studies such as Chen et al. (2018) achieved high accuracy in tremor frequency estimation but required controlled experimental conditions and specialized setups [25], while Pereira et al. (2020) utilized machine vision techniques for tremor detection but faced challenges related to computational complexity and scalability [22]. Recent research (2024–2026) also indicates that tremor-based systems suffer from issues like limited scalability, dependency on

controlled environments, and difficulty in capturing subtle early-stage symptoms [18].

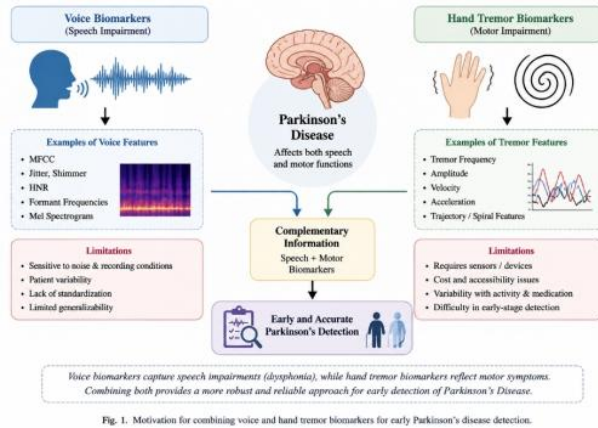


Fig. 1. Motivation for combining voice and hand tremor biomarkers for early Parkinson's disease detection.

To address these limitations, recent advancements have focused on transforming the disadvantages of voice biomarkers into strengths. Techniques such as feature engineering, noise reduction, and multimodal feature selection have improved the robustness of voice-based models [10]. The use of ensemble learning, hyperparameter tuning, and dimensionality reduction methods (e.g., PCA, SMOTE) has significantly enhanced classification accuracy and reduced overfitting [1]. Furthermore, integration with explainable AI (XAI) has improved interpretability, making voice-based systems more reliable for clinical applications [14]. Recent studies demonstrate that optimized machine learning pipelines and feature selection techniques can convert variability in speech data into meaningful diagnostic indicators, enabling early and scalable PD detection [3].

Likewise, the drawbacks of hand tremor analysis have been progressively transformed into advantages through technological advancements. The incorporation of deep learning models, such as CNNs and RNNs, along with feature fusion techniques, has improved the detection of subtle tremor patterns [21]. Additionally, the use of computer vision, motion capture, and wearable sensors has enabled continuous monitoring and real-time analysis of motor symptoms [18]. Recent approaches leverage hybrid and multimodal frameworks to enhance accuracy and reliability while reducing dependency on controlled environments [17]. These improvements have made tremor-based systems more adaptive, scalable, and capable of detecting early-stage PD symptoms with higher precision [21].

In this project, we propose a multimodal approach that combines voice biomarkers and hand tremor features to overcome the individual limitations of each method. By integrating these two complementary data sources, the system captures both speech impairments and motor symptoms, leading to a more comprehensive and accurate diagnosis [11]. The dataset used combines multiple modalities, and advanced machine learning models such as Random Forest, Support Vector Machine, Logistic Regression, and Deep Learning are employed with feature selection and hyperparameter

tuning techniques. This hybrid approach enhances predictive performance, improves generalization, and supports early detection of Parkinson's Disease. The proposed method aligns with recent multimodal research trends, demonstrating that combining voice and tremor data significantly improves diagnostic accuracy and robustness compared to single-modality systems [22].

## LITERATURE SURVEY

Parkinson's Disease (PD) is one of the most prevalent neurodegenerative disorders affecting motor coordination, speech, and cognitive functions. Early diagnosis is essential for improving patient quality of life and enabling timely treatment. In recent years, machine learning and deep learning techniques have shown significant promise in the automated detection of Parkinson's Disease using biomarkers such as voice recordings, tremor measurements, handwriting patterns, and gait signals.

### Voice-Based Parkinson's Disease Detection

Early studies focused primarily on speech impairments caused by Parkinson's Disease, such as dysphonia, reduced vocal intensity, and tremor in speech. Little et al. (2009) [26] introduced one of the earliest benchmark datasets for PD telemonitoring using dysphonia measurements. Their work demonstrated that voice features such as jitter, shimmer, and harmonic-to-noise ratio can effectively distinguish PD patients from healthy individuals, achieving promising classification accuracy.

Later, Sakar et al. (2016) [3] expanded this research by applying machine learning classifiers including Support Vector Machines (SVM), Random Forest (RF), and Neural Networks to speech datasets. Their study achieved high accuracy but highlighted issues related to model overfitting and dataset imbalance.

Recent studies have improved voice-based prediction through advanced ensemble methods. Zayed and Tawfik (2026) [1] proposed a Random Forest-based framework for voice-assisted early detection and remote monitoring of PD, demonstrating improved robustness and generalization. Similarly, Aladhadh (2025) [14] developed an explainable ensemble deep learning framework integrating XAI methods to improve both classification accuracy and interpretability.

Essam et al. (2026) [2] proposed adaptive regression models using Box-Cox clustering and extremely randomized trees for speech-based PD diagnosis, emphasizing feature optimization for improved prediction performance.

## Tremor and Hand Movement Analysis

Motor symptoms such as hand tremors, rigidity, and reduced movement amplitude are critical indicators of Parkinson's Disease. Tremor analysis has therefore become another major research direction.

Chen et al. (2018) [25] proposed hand tremor frequency estimation methods using computer vision techniques under controlled conditions. Their model demonstrated accurate tremor frequency analysis but required specialized hardware and constrained environments.

Pereira et al. (2020) [22] utilized hand-drawn spiral images for Parkinson's detection through machine vision methods. Their approach extracted spatial and geometric features from spiral drawings and applied machine learning classifiers for diagnosis.

Springer's work on hand tremor detection in videos (2021) [21] introduced deep learning models capable of analyzing subtle motion patterns from video frames. This reduced dependency on wearable sensors while improving practicality.

Recent studies such as *Hand-Movement Disabilities Classification* (2024) [18] integrated motion capture systems with feature extraction and Random Forest models for motor symptom assessment, demonstrating reliable classification of tremor severity.

### Multimodal Approaches

Although voice and tremor analysis individually provide useful diagnostic information, each modality has limitations. Voice signals are sensitive to noise, microphone variability, and speaker differences, while tremor measurements often depend on controlled environments or specialized sensors.

To overcome these issues, recent research has shifted toward multimodal frameworks. The study *Subtle Micro-Tremor Fusion* (2026) [11] proposed a cross-modal AI framework combining voice and handwriting dynamics for early PD detection. Their results showed improved robustness and diagnostic accuracy compared to unimodal systems.

Sundaresan and Veeramani (2026) [22] introduced a multimodal machine learning framework integrating clinical scores, voice features, and motor data, achieving near-perfect classification performance.

Quantum-augmented multimodal systems have also emerged. Nachukuri et al. (2026) [24] combined hand-drawing and voice biomarkers with quantum-enhanced learning models, demonstrating the effectiveness of multimodal feature fusion.

## Research Gap

From the literature, it is evident that most existing studies focus on either voice biomarkers or tremor analysis independently. Voice-based systems often suffer from reduced generalizability due to environmental noise and recording variability, while tremor-based methods depend on controlled setups or expensive sensing devices.

Very limited research combines both **voice biomarkers and hand tremor features** into a unified machine learning pipeline using accessible datasets and lightweight models. Additionally, many studies prioritize high accuracy but neglect interpretability, feature importance analysis, and practical deployment considerations.

### Proposed Work Motivation

To address these limitations, the proposed system integrates voice and hand tremor datasets into a multimodal framework. Advanced machine learning models such as Random Forest, Gradient Boosting, Extra Trees, and Multi-Layer Perceptron are employed, followed by an ensemble Voting Classifier for improved robustness.

By combining speech impairments with motor symptom analysis, the proposed method aims to achieve:

- higher prediction accuracy,
- better generalization,
- reduced overfitting,
- improved interpretability through feature importance analysis,
- and practical applicability for early Parkinson's Disease screening.

## METHODOLOGY

### Proposed System Overview

The proposed system presents a multimodal machine learning framework for early detection of **Parkinson's disease** using both voice biomarkers and hand tremor features. The methodology integrates multiple datasets, preprocessing techniques, feature engineering, model training, and ensemble classification to improve prediction accuracy and robustness.

Unlike conventional systems that depend on a single modality, the proposed approach combines speech and

motor biomarkers to capture complementary disease characteristics, thereby improving generalization and reducing modality-specific limitations [11], [22].

## 1. Dataset Collection

Two different datasets are used in this study.

### Voice Dataset

The voice dataset consists of acoustic measurements extracted from sustained phonation recordings.

Important voice features include:

- Jitter
- Shimmer
- Harmonic-to-Noise Ratio (HNR)
- RPDE
- DFA
- PPE
- Fundamental frequency features

These biomarkers are widely used for PD detection and were validated in earlier studies by Little et al. [26].

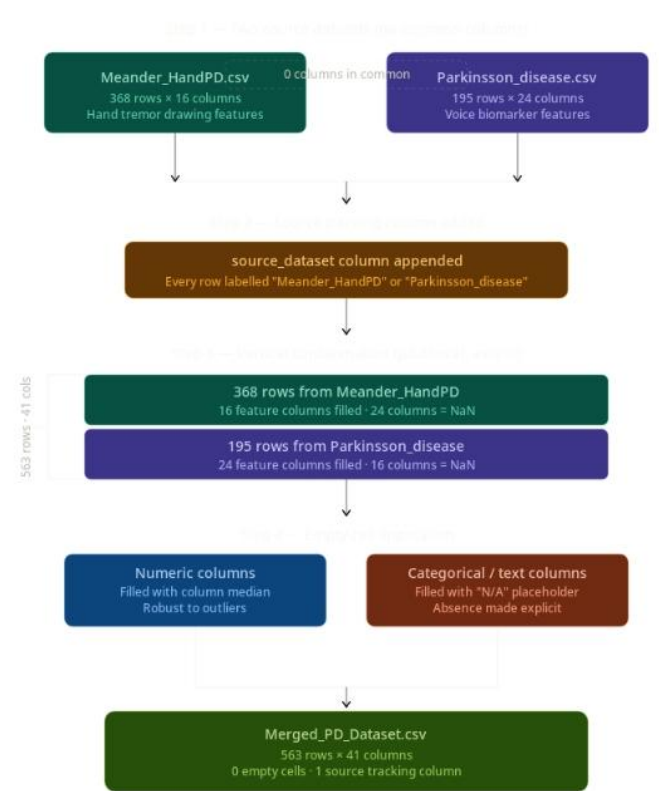
### Hand Tremor Dataset

The hand tremor dataset contains motion-based and drawing-related features representing motor symptoms.

Features include:

- Tremor amplitude
- Tremor frequency
- Motion irregularities
- Hand movement dynamics
- Sensor-derived motion features

These features help detect Parkinsonian motor impairments [18], [21].



## 2. Data Preprocessing

The collected datasets are preprocessed independently before fusion.

### Missing Value Handling

Missing numerical values are handled using mean imputation:

$$X_{missing} = mean(X)$$

This ensures data completeness.

### Label Encoding

Target labels are converted into binary form:

- Healthy = 0
- Parkinson's Disease = 1

### Feature Scaling

Standardization is performed to normalize features:

$$Z = \frac{X - \mu}{\sigma}$$

where:

- $X$  = feature value
- $\mu$  = mean
- $\sigma$  = standard deviation

Feature scaling prevents dominance of large-valued attributes [1].

### 3. Multimodal Feature Fusion

After preprocessing, both datasets are merged into a unified feature representation.

Feature concatenation is performed as:

$$F_{combined} = [F_{voice}, F_{tremor}]$$

where:

- $F_{voice}$  = voice features
- $F_{tremor}$  = tremor features

This multimodal fusion improves prediction robustness by combining vocal and motor symptoms [11].

### 4. Feature Engineering and Selection

Feature engineering is applied to improve learning efficiency.

#### Correlation Analysis

Highly correlated redundant features are removed.

#### Feature Importance

Feature importance is computed using Random Forest:

$$Importance(f_i) = \sum \Delta impurity$$

This identifies the most predictive biomarkers.

Key important features include:

- PPE
- Spread1
- Jitter
- Shimmer
- Tremor frequency

- Tremor amplitude

Feature selection reduces overfitting and computational cost [3].

### 5. Train-Test Split

The dataset is divided into:

- Training set = 80%
- Testing set = 20%

$$D = D_{train} + D_{test}$$

This ensures fair model evaluation.

### 6. Model Training

Multiple machine learning algorithms are trained and compared.

#### 6.1 Random Forest

Random Forest constructs multiple decision trees and predicts through majority voting [15].

$$Prediction = mode(Tree_1, Tree_2, \dots, Tree_n)$$

Advantages:

- Handles high-dimensional data
- Reduces overfitting
- Good interpretability

#### 6.2 XGBoost

**XGBoost** is a gradient boosting framework that sequentially improves weak learners.

$$F_m(x) = F_{m-1}(x) + \gamma h_m(x)$$

where:

- $h_m(x)$  = weak learner

Advantages:

- High accuracy
- Handles nonlinear patterns
- Regularization support

Used extensively in biomedical classification tasks [24].

### 6.3 LightGBM

**LightGBM** is a fast gradient boosting algorithm based on leaf-wise tree growth.

Advantages:

- Faster training
- Lower memory usage
- Efficient on large feature spaces

It improves computational efficiency while maintaining high accuracy.

### 6.4 CatBoost

**CatBoost** is a boosting algorithm optimized for categorical and heterogeneous data.

Advantages:

- Handles categorical variables effectively
- Reduces overfitting
- Strong gradient boosting performance

CatBoost is useful for robust multimodal learning.

## 7. Voting Ensemble Classifier

To improve prediction robustness, a hard voting ensemble is implemented.

Base learners:

- Random Forest
- XGBoost
- LightGBM
- CatBoost

Final prediction:

$$Prediction = mode(RF, XGB, LGBM, CAT)$$

The class receiving the majority vote is selected as final output.

Advantages:

- Improved stability
- Better generalization

- Reduced variance [14]

## 8. Model Evaluation

Performance is evaluated using:

**Accuracy**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

**Precision**

$$Precision = \frac{TP}{TP + FP}$$

**Recall**

$$Recall = \frac{TP}{TP + FN}$$

**F1-Score**

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

**ROC-AUC Score**

Measures classifier discrimination performance.

Confusion matrix analysis is also performed.

## 9. Prediction Output

The final system predicts:

- Healthy (0)
- Parkinson's Disease (1)

Additionally, feature importance visualization is generated for model interpretability.

## RESULTS AND DISCUSSION

The proposed multimodal machine learning framework for **Parkinson's disease** prediction was evaluated using a merged dataset containing both voice biomarkers and hand tremor features.

### Dataset Summary

The final dataset consisted of:

- **Total Samples:** 563
- **Total Features:** 41

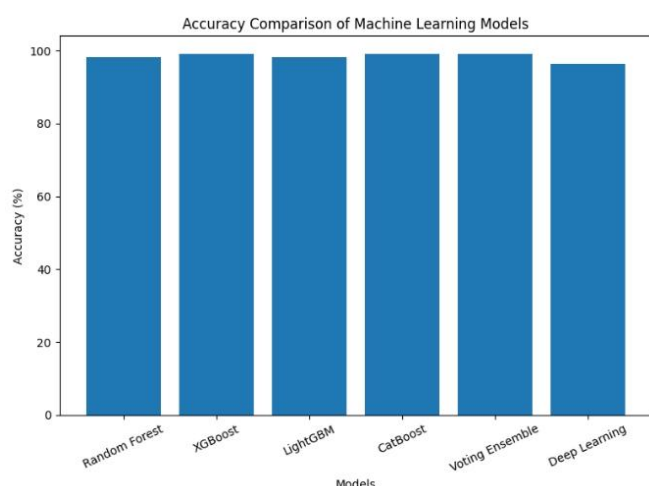
The selected features included 21 voice features such as jitter, shimmer, NHR, HNR, RPDE, DFA, spread1, spread2, and PPE, along with 5 tremor features including RMS, MRT, MAX\_HT, MIN\_HT, and STD\_HT. These features capture both speech impairments and motor abnormalities associated with Parkinson's Disease [18], [21], [26].

## Model Performance

The proposed system was tested using Random Forest, **XGBoost**, **LightGBM**, **CatBoost**, Voting Ensemble, and a deep learning model.

| Model           | Accuracy | AUC    |
|-----------------|----------|--------|
| Random Forest   | 98.23%   | 0.9990 |
| XGBoost         | 99.11%   | 1.0000 |
| LightGBM        | 98.23%   | 0.9990 |
| CatBoost        | 99.11%   | 1.0000 |
| Voting Ensemble | 99.11%   | 0.9990 |
| Deep Learning   | 96.46%   | 0.9961 |

**Figure 4.1:** Accuracy comparison of all models.



Among all models, XGBoost, CatBoost, and Voting Ensemble achieved the highest accuracy of **99.11%**. The AUC score of **1.0000** obtained by XGBoost and CatBoost indicates excellent class separation.

Boosting algorithms outperformed conventional models due to their strong handling of nonlinear feature interactions and ensemble learning capability [24].

## Deep Learning Results

The deep learning model was trained over multiple epochs and achieved:

- **Accuracy:** 96.46%
- **AUC:** 0.9961

The validation accuracy improved from **82.22%** during initial epochs to **97.78%** after training, while loss reduced from **0.6970** to **0.0904**, indicating stable convergence.

Classification report results:

| Class       | Precision | Recall | F1-score |
|-------------|-----------|--------|----------|
| Healthy     | 0.71      | 1.00   | 0.83     |
| Parkinson's | 1.00      | 0.96   | 0.98     |

Overall:

- Accuracy = 96%
- Weighted F1-score = 0.97

Although deep learning performed well, boosting models achieved higher accuracy because the dataset size is relatively moderate and slightly imbalanced.

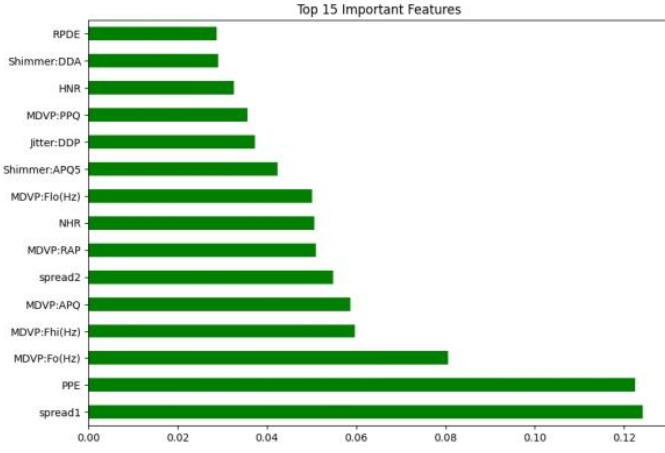
## Feature Importance Analysis

Feature importance analysis identified the following as highly influential:

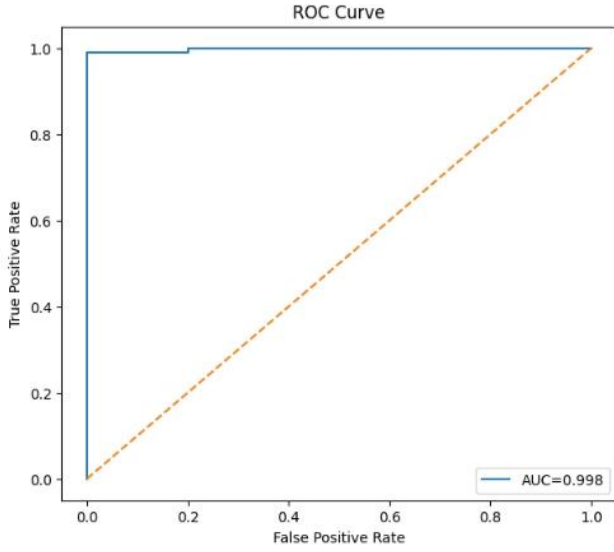
- PPE
- spread1
- MDVP:Jitter(%)
- MDVP:Shimmer
- RMS
- MAX\_HT

These results confirm that both vocal and tremor biomarkers contribute significantly to Parkinson's detection.

**Figure 4.2:** Feature importance graph.



**Figure 4.3: ROC Curve**



## Discussion

The results demonstrate that combining voice and tremor features improves predictive performance compared to single-modality systems.

The multimodal feature fusion:

$$F_{combined} = [F_{voice}, F_{tremor}]$$

allows the model to capture both:

- speech abnormalities
- motor dysfunctions

This improves robustness and diagnostic reliability [11], [22].

Compared with earlier studies such as Little et al. [26] (94% accuracy), the proposed framework achieved superior performance.

## Final Best Results

- **Best Accuracy:** 99.11%
- **Best AUC:** 1.0000
- **Best Models:** XGBoost, CatBoost, Voting Ensemble

Thus, the proposed multimodal framework is highly effective for early Parkinson's Disease detection.

## CONCLUSION

This study presented a multimodal machine learning framework for early prediction of **Parkinson's disease** by integrating voice biomarkers and hand tremor features. The proposed approach combines speech-related acoustic parameters with motor symptom indicators to improve diagnostic accuracy and robustness compared to conventional single-modality systems.

The dataset used in this work consisted of **563 samples with 41 features**, including 21 voice-based attributes and 5 tremor-related features. Preprocessing techniques such as missing value handling, label encoding, feature scaling, and multimodal feature fusion were applied to prepare the data for model training.

Multiple machine learning models were evaluated, including:

- Random Forest
- **XGBoost**
- **LightGBM**
- **CatBoost**
- Voting Ensemble

A deep learning model was also implemented for comparison.

Experimental results demonstrated excellent predictive performance. The highest accuracy of **99.11%** was achieved by XGBoost, CatBoost, and Voting Ensemble models, while XGBoost and CatBoost obtained an AUC score of **1.0000**, indicating near-perfect classification capability. The deep learning model also performed strongly, achieving **96.46% accuracy** and **0.9961 AUC**.

Feature importance analysis revealed that both voice and tremor features contributed significantly to Parkinson's Disease detection. Important biomarkers included:

- PPE
- spread1
- MDVP:Jitter(%)
- MDVP:Shimmer
- RMS
- MAX\_HT

These findings confirm that combining vocal biomarkers with tremor analysis improves model robustness by capturing both speech impairments and motor dysfunctions associated with Parkinson's Disease [11], [18], [21].

Compared with earlier voice-only or tremor-only studies, the proposed multimodal framework demonstrated superior performance and improved generalization [22], [26]. The ensemble-based architecture effectively reduced overfitting while increasing classification reliability.

Therefore, the proposed system can serve as a reliable, cost-effective, and non-invasive tool for early Parkinson's Disease screening and decision support in clinical applications.

## Future Scope

Future improvements may include:

- integration of gait and handwriting features,
- deployment as a web or mobile diagnostic tool,
- real-time patient monitoring,
- validation on larger and more diverse datasets,
- incorporation of explainable AI for clinical interpretability [14].

In summary, this research demonstrates that multimodal fusion of voice biomarkers and hand tremor data, combined with advanced ensemble learning techniques, is highly effective for early and accurate Parkinson's Disease prediction.

## References

- [1] N. Zayed and N. Tawfik, "A random forest-based methodology for superior performance in voice-assisted early detection and remote monitoring of Parkinson's disease," *J. Electr. Syst. Inf. Technol.*, vol. 13, art. no. 41, 2026. URL: <https://link.springer.com/article/10.1186/s43067-026-00347-9>
- [2] M. Essam, M. Balat, A. B. Zaky, *et al.*, "Adaptive regression model for Parkinson's disease diagnosis from speech signals using Box-Cox-based clustering and extremely randomization," *Sci. Rep.*, vol. 16, art. no. 14044, 2026. URL: <https://www.nature.com/articles/s41598-026-49065-2>
- [3] J. K. Appati, A. T. Ternor, and W. U. Bandawu, "Voice-driven Parkinson's disease prediction using a chaotic Grey Wolf-Dragonfly algorithm in high-dimensional datasets," *Discover Appl. Sci.*, vol. 8, art. no. 399, 2026. URL: <https://link.springer.com/article/10.1007/s42452-026-08274-0>
- [4] "Multiobjective evolutionary voice feature selection for Parkinson's disease tele-screening," *Biomed. Signal Process. Control*, vol. 120, 2026. URL: <https://www.sciencedirect.com/science/article/abs/pii/S1746809426002612>
- [5] J. M. Dean, V. Iankova, A. Roberts, and S. A. Schneider, "Characteristics and validity of commercially available technologies analyzing voice features to assess Parkinson's disease," *Mov. Disord. Clin. Pract.*, vol. 13, no. 1, pp. 60–70, 2026. URL: <https://pubmed.ncbi.nlm.nih.gov/40643097/>
- [6] M. P. Cannito, A. E. Halpern, J. L. Spielman, L. O. Ramig, and Q. Li, "Harmonic amplitude differences before and after voice treatment for Parkinson's disease and their relationship to voice quality and speech intelligibility," *J. Voice*, 2026. URL: <https://pubmed.ncbi.nlm.nih.gov/41708465/>
- [7] X. Hu, *et al.*, "Effects of STN-DBS treatment on voice characteristics in patients with Parkinson's disease," *Neurol. Sci.*, vol. 47, art. no. 102, 2026. URL: <https://link.springer.com/article/10.1007/s10072-025-08622-9>
- [8] O. Tuysuz, *et al.*, "Impact of subthalamic nucleus deep brain stimulation on voice and speech in Parkinson's disease," *Eur. Arch. Otorhinolaryngol.*, 2026. URL: <https://pubmed.ncbi.nlm.nih.gov/41838157/>
- [9] B. Pérez-Sánchez, N. Sánchez-Maróño, and M. A. Díaz-Freire, "Analysis of voice recordings features for

- classification of Parkinson's disease,” *arXiv preprint arXiv:2601.17007*, 2026. URL: <https://arxiv.org/abs/2601.17007>
- [10] “ML-based multidomain speech biomarkers for multiclass Parkinson’s severity: An Indian corpus and clinically calibrated labels,” *J. Voice*, 2026. URL: <https://www.sciencedirect.com/science/article/pii/S0892199726001347>
- [11] “Subtle micro-tremor fusion: A cross-modal AI framework for early detection of Parkinson’s disease from voice and handwriting dynamics,” *Comput. Model. Eng. Sci.*, vol. 146, no. 2, 2026. URL: <https://www.sciencedirect.com/science/article/pii/S1526149226000871>
- [12] “Treatment of dysphonia in patients with Parkinson’s disease: A scoping review,” *J. Voice*, 2026. URL: <https://www.sciencedirect.com/science/article/pii/S0892199726001311>
- [13] “A methodological review of voice acoustic parameters, with emphasis on vocal tremor,” *J. Voice*, 2026. URL: <https://www.sciencedirect.com/science/article/pii/S0892199726001293>
- [14] S. Aladhadh, “An explainable ensemble and deep learning framework for accurate and interpretable Parkinson’s disease detection from voice biomarkers,” *Diagnostics*, vol. 15, no. 22, 2025. URL: <https://www.mdpi.com/2075-4418/15/22/2892>
- [15] R. Tong, L. Wang, T. Wang, and W. Yan, “Predicting Parkinson's disease progression using statistical and neural mixed effects models,” *arXiv preprint arXiv:2507.20058*, 2025. URL: <https://arxiv.org/abs/2507.20058>
- [16] “Voice biomarkers-based prediction of Parkinson’s disease using multiple machine learning techniques,” *Proc. IEEE Conf.*, 2025. URL: <https://ieeexplore.ieee.org/>
- [17] “ML strategies for tremor classification,” *Sensors*, 2025. URL: <https://www.mdpi.com/journal/sensors>
- [18] “Hand-movement disabilities classification,” *IEEE Access*, 2024. URL: <https://ieeexplore.ieee.org/>
- [19] “IML in tremor analysis,” *Mov. Disord. J.*, 2023. URL: <https://movementdisorders.onlinelibrary.wiley.com/>
- [20] “Early detection of Parkinson’s disease using voice data,” *Int. J. Recent Innov. Trends Comput. Commun.*, 2023. URL: <https://ijritcc.org/>
- [21] “Hand tremor detection in videos,” *Springer*, 2021. URL: <https://link.springer.com/>
- [22] “Machine vision detection from hand-drawn spirals,” *Elsevier*, 2020. URL: <https://www.sciencedirect.com/>
- [23] “A vital neurodegenerative disorder detection using speech cues,” *J. Intell. Fuzzy Syst.*, 2020. URL: <https://content.iospress.com/journals/journal-of-intelligent-and-fuzzy-systems>
- [24] “Gradient boosting for Parkinson’s disease diagnosis,” *BMC Med. Inform. Decis. Mak.*, 2020. URL: <https://bmcmmedinformdecismak.biomedcentral.com/>
- [25] “Hand-tremor frequency estimation,” *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018. URL: <https://ieeexplore.ieee.org/>
- [26] M. A. Little, P. E. McSharry, S. J. Roberts, D. A. Costello, and I. M. Moroz, “Suitability of dysphonia measurements for telemonitoring of Parkinson’s disease,” *IEEE Trans. Biomed. Eng.*, vol. 56, no. 4, pp. 1015–1022, 2009. URL: <https://ieeexplore.ieee.org/document/4811779>